

Lab Guide 10

Manual And Automatic Scaling

Q.1 Create one deployment using nginx image and scale it manually to 5 pods

Q.2 Create one dep and scaling up or down should happen automatically with respect to CPU usage.

Q.3 Create one deployment and using VPA get recommended value of resources for deployment.

Ans: 1

```
kubectl create deployment webserver --image nginx
```

```
kubectl get all
```

```
kubectl get pods
```

```
kubectl scale deployment webserver --replicas=5
```

```
kubectl get pods
```

```

root@master:~# kubectl create deployment webserver --image nginx
deployment.apps/webserver created
root@master:~# kubectl get all
NAME                                READY   STATUS    RESTARTS   AGE
pod/pod                             1/1     Running   2           20h
pod/webserver-667ddc69b6-24wvr      1/1     Running   0           3s

NAME                                TYPE               CLUSTER-IP   EXTERNAL-IP   PORT(S)          AGE
service/kubernetes                  ClusterIP        10.96.0.1    <none>        443/TCP          20h
service/pod                         NodePort         10.96.34.95  <none>        80:30326/TCP     20h

NAME                                READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/webserver            1/1     1             1           3s

NAME                                DESIRED   CURRENT   READY   AGE
replicaset.apps/webserver-667ddc69b6 1         1         1       3s
root@master:~# kubectl get [pds
error: the server doesn't have a resource type "[pds"
root@master:~# kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
pod                                 1/1     Running   2           20h
webserver-667ddc69b6-24wvr         1/1     Running   0           7s
root@master:~# kubectl scale deployment webserver --replicas=5
deployment.apps/webserver scaled
root@master:~# kubectl get pods
NAME                                READY   STATUS             RESTARTS   AGE
pod                                 1/1     Running            2           20h
webserver-667ddc69b6-24wvr         1/1     Running            0           24s
webserver-667ddc69b6-6pvct         1/1     Running            0           2s
webserver-667ddc69b6-cnr2b         0/1     ContainerCreating   0           2s
webserver-667ddc69b6-hprp9         0/1     ContainerCreating   0           2s
webserver-667ddc69b6-z79kt         0/1     ContainerCreating   0           2s
root@master:~#

```

Ans.2

Create dephpa.yml file

vim dephpa.yaml

```

apiVersion: apps/v1
kind: Deployment
metadata:
  name: php-apache
spec:
  selector:
    matchLabels:
      run: php-apache
  replicas: 1
  template:
    metadata:
      labels:
        run: php-apache
    spec:

```

```

    containers:
      - name: php-apache
        image: k8s.gcr.io/hpa-example
        ports:
          - containerPort: 80
        resources:
          limits:
            cpu: 500m
          requests:
            cpu: 200m
---
apiVersion: v1
kind: Service
metadata:
  name: php-apache
  labels:
    run: php-apache
spec:
  ports:
    - port: 80
  selector:
    run: php-apache

```

kubectl create -f dephpa.yaml

kubectl get all

kubectl top node

kubectl autoscale deployment php-apache --cpu-percent=50 --min=1 --max=10

kubectl run -i --tty load-generator --rm --image=busybox:1.28 --restart=Never -- /bin/sh -c "while sleep 0.01; do wget -q -O- http://php-apache; done"

kubectl get hpa php-apache --watch

Ans. 3

Step : 1

Install VPA first

git clone <https://github.com/kubernetes/autoscaler.git>

cd autoscaler/vertical-pod-autoscaler/hack/

ls

bash vpa-up.sh

it will take some time and after that you can verify by running below command

kubectl get vpa

```
root@master:~/autoscaler/vertical-pod-autoscaler/hack# kubectl get vpa
No resources found in default namespace.
root@master:~/autoscaler/vertical-pod-autoscaler/hack#
```

Step 2

Create dep

Cat vpa-dep.yml

apiVersion: apps/v1

kind: Deployment

metadata:

```
name: nginx-deployment
```

```
spec:
```

```
replicas: 1
```

```
selector:
```

```
matchLabels:
```

```
  app: nginx
```

```
template:
```

```
metadata:
```

```
  labels:
```

```
    app: nginx
```

```
spec:
```

```
containers:
```

```
- name: nginx
```

```
  image: nginx
```

```
kubectl create -f vpa-dep.yml
```

```
root@master:~# cat vpa-dep.yml
# nginx-deployment.yml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
spec:
  replicas: 1
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
        - name: nginx
          image: nginx

root@master:~# kubectl create -f vpa-dep.yml
deployment.apps/nginx-deployment created
```

Step 3

Create vpa yaml

```
# vpa.yaml
apiVersion: autoscaling.k8s.io/v1
kind: VerticalPodAutoscaler
metadata:
  name: nginx-vpa
```

spec:

targetRef:

apiVersion: "apps/v1"

kind: Deployment

name: nginx-deployment

updatePolicy:

updateMode: "Auto"

kubectl create -f vpa.yml

verticalpodautoscaler.autoscaling.k8s.io/nginx-vpa created

```
root@master:~# cat vpa.yml
# vpa.yml
apiVersion: autoscaling.k8s.io/v1
kind: VerticalPodAutoscaler
metadata:
  name: nginx-vpa
spec:
  targetRef:
    apiVersion: "apps/v1"
    kind: Deployment
    name: nginx-deployment
  updatePolicy:
    updateMode: "Auto"

root@master:~# kubectl create -f vpa.yml
Warning: UpdateMode "Auto" is deprecated and will be removed in a future API version. Use explicit update modes like "Recreate",
"Initial", or "InPlaceOrRecreate" instead. See https://github.com/kubernetes/autoscaler/issues/8424 for more details.
verticalpodautoscaler.autoscaling.k8s.io/nginx-vpa created
```

Step 4 describe VPA for details

kubectl describe vpa nginx-vpa

```
Annotations:  <none>
API Version:  autoscaling.k8s.io/v1
Kind:         VerticalPodAutoscaler
Metadata:
  Creation Timestamp:  2025-09-28T05:16:57Z
  Generation:         1
  Resource Version:    1925
  UID:                f0204225-264a-4f92-a4f1-b473915b58f3
Spec:
  Target Ref:
    API Version:  apps/v1
    Kind:         Deployment
    Name:         nginx-deployment
  Update Policy:
    Update Mode:  Auto
Status:
  Conditions:
    Last Transition Time:  2025-09-28T05:17:45Z
    Status:               True
    Type:                 RecommendationProvided
  Recommendation:
    Container Recommendations:
      Container Name:  nginx
      Lower Bound:
        Cpu:          25m
        Memory:       250Mi
      Target:
        Cpu:          25m
        Memory:       250Mi
      Uncapped Target:
        Cpu:          25m
        Memory:       250Mi
      Upper Bound:
        Cpu:          25m
        Memory:       250Mi
Events:  <none>
```